

HDOC Day-35 — String Operations Using Functions

1. Algorithm

Step 1: Start the program.

Step 2: Declare a character array to store the string and an integer `choice`.

Step 3: Read a string from the user using `fgets()`.

Step 4: Display the menu:

1. Count characters in the string.
2. Print each character on a new line.
3. Count vowels and consonants.
4. Exit.

Step 5: Read the user's choice.

Step 6: Use a `switch` statement:

- **Choice 1:** Call `countCharacters()` to count characters without using `strlen()`.
- **Choice 2:** Call `printCharacters()` to print every character on a separate line.
- **Choice 3:** Call `countVowelsConsonants()` to count vowels and consonants.
- **Choice 4:** Exit the program.
- **Default:** Display "Invalid choice!".

Step 7: Continue displaying the menu until the user selects option 4.

Step 8: Stop.

2. C Program

```
#include <stdio.h>

int countCharacters(char str[])
{
    int i = 0;

    while (str[i] != '\0')
    {
        i++;
    }

    return i;
}
```

```

void printCharacters(char str[])
{
    int i = 0;

    while (str[i] != '\0')
    {
        printf("%c\n", str[i]);
        i++;
    }
}

void countVowelsConsonants(char str[])
{
    int i = 0;
    int vowels = 0, consonants = 0;

    while (str[i] != '\0')
    {
        if (str[i] == 'a' || str[i] == 'e' ||
            str[i] == 'i' || str[i] == 'o' ||
            str[i] == 'u' || str[i] == 'A' ||
            str[i] == 'E' || str[i] == 'I' ||
            str[i] == 'O' || str[i] == 'U')
        {
            vowels++;
        }
        else if ((str[i] >= 'a' && str[i] <= 'z') ||
            (str[i] >= 'A' && str[i] <= 'Z'))
        {
            consonants++;
        }

        i++;
    }

    printf("Number of vowels = %d\n", vowels);
    printf("Number of consonants = %d\n", consonants);
}

int main()
{
    char str[100];
    int choice;

    printf("Enter a string: ");
    scanf(" %[^\\n]", str);

    do
    {
        printf("\n--- MENU ---\n");
        printf("1. Count characters in the string\n");
        printf("2. Print each character on a new line\n");
        printf("3. Count vowels and consonants\n");
        printf("4. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
    }
}

```

```

switch (choice)
{
    case 1:
        printf("Number of characters = %d\n",
            countCharacters(str));
        break;

    case 2:
        printf("Characters of the string are:\n");
        printCharacters(str);
        break;

    case 3:
        countVowelsConsonants(str);
        break;

    case 4:
        printf("Exiting the program...\n");
        break;

    default:
        printf("Invalid choice! Please enter 1 to 4.\n");
}

} while (choice != 4);

return 0;
}

```

3. Test Case Table

Test Case	Input String	Choice	Expected Output
TC-1	Hello	1	Number of characters = 5
TC-2	Code	2	C, o, d, e printed on separate lines
TC-3	Hello	3	Vowels = 2, Consonants = 3
TC-4	AEIOU	3	Vowels = 5, Consonants = 0
TC-5	C Programming	3	Vowels = 3, Consonants = 9
TC-6	Hello World	1	Number of characters = 11
TC-7	123abc	3	Vowels = 1, Consonants = 2
TC-8	Hello	5	Invalid choice
TC-9	Hello	4	Program exits

4. Test Case Simulation

Test Case 1 — Count Characters

```
Enter a string: Hello

— MENU —
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 1
Number of characters = 5
```

Evaluation:

The loop checks each character until `'\0'` is encountered. There are five characters in `Hello`, so the program correctly displays 5.

Test Case 2 — Print Each Character

```
Enter a string: Code

— MENU —
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 2
Characters of the string are:
C
o
d
e
```

Evaluation:

The `printCharacters()` function traverses the string one character at a time. Each character is printed using `\n`, so every character appears on a separate line.

Test Case 3 — Count Vowels and Consonants

```
Enter a string: Hello

— MENU —
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Number of vowels = 2
Number of consonants = 3
```

Evaluation:

In `Hello`, the vowels are **e, o**, giving 2 vowels. The consonants are **H, l, l**, giving 3 consonants. Therefore, the output is correct.

Test Case 4 — All Vowels

```
Enter a string: AEIOU

— MENU —
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Number of vowels = 5
Number of consonants = 0
```

Evaluation:

All five characters are uppercase vowels. The program checks both uppercase and lowercase vowels, so it correctly counts **5 vowels and 0 consonants**.

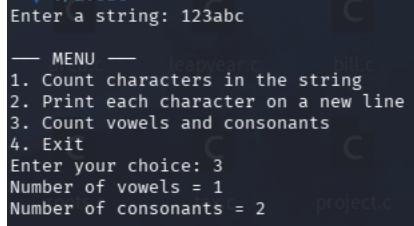
Test Case 5 — String Containing a Space

```
Enter a string: C Programming

— MENU —
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Number of vowels = 3
Number of consonants = 9
```

Evaluation:

The vowels are **o**, **a**, **i**, while the remaining alphabetic characters are consonants. The space is ignored while counting vowels and consonants.

Test Case 6 — Numbers and Letters

```
Enter a string: 123abc
— MENU —
1. Count characters in the string
2. Print each character on a new line
3. Count vowels and consonants
4. Exit
Enter your choice: 3
Number of vowels = 1
Number of consonants = 2
```

Evaluation:

The vowel is **a**, and the consonants are **b** and **c**. The digits **1**, **2**, and **3** are neither vowels nor consonants, so they are ignored.
